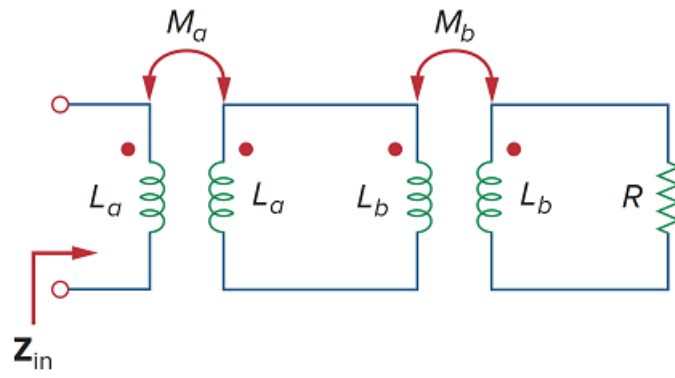


Problem

Two linear transformers are cascaded as shown in Fig. 13.101. Show that

$$Z_{in} = \frac{\omega^2 R (L_a^2 + L_a L_b - M_a^2) + j\omega^3 (L_a^2 L_b + L_a L_b^2 - L_a M_b^2 - L_b M_a^2)}{\omega^2 (L_a L_b + L_b^2 - M_b^2) - j\omega R (L_a + L_b)}$$

Figure 13.101

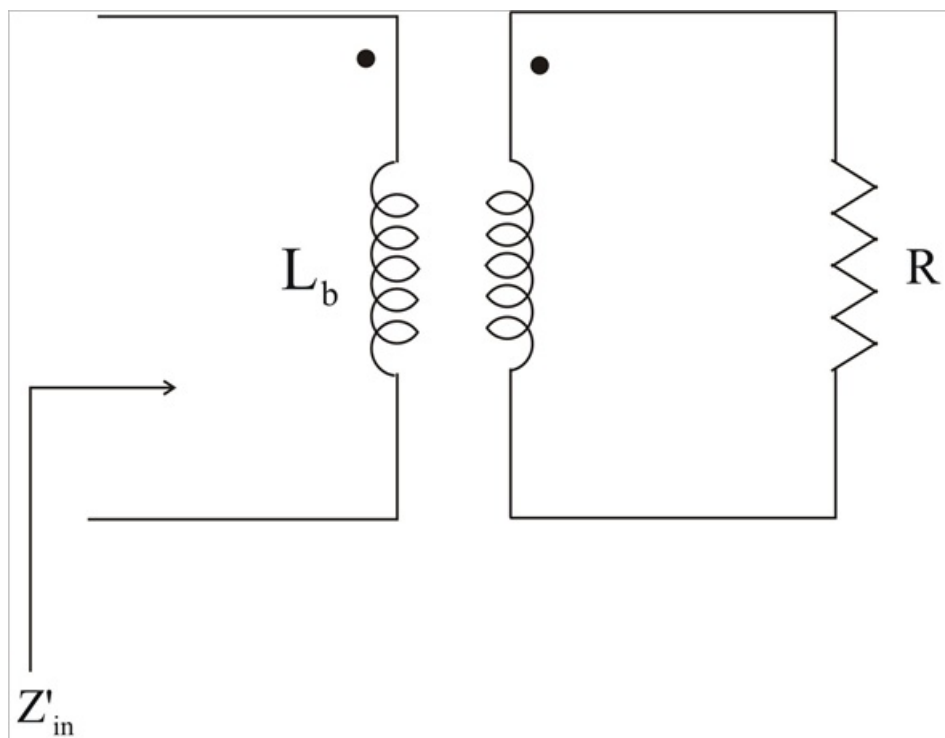


Step-by-step solution

Step 1 of 5

We first find Z_{in} for the second stage using the concept of reflected impedance.

Step 2 of 5

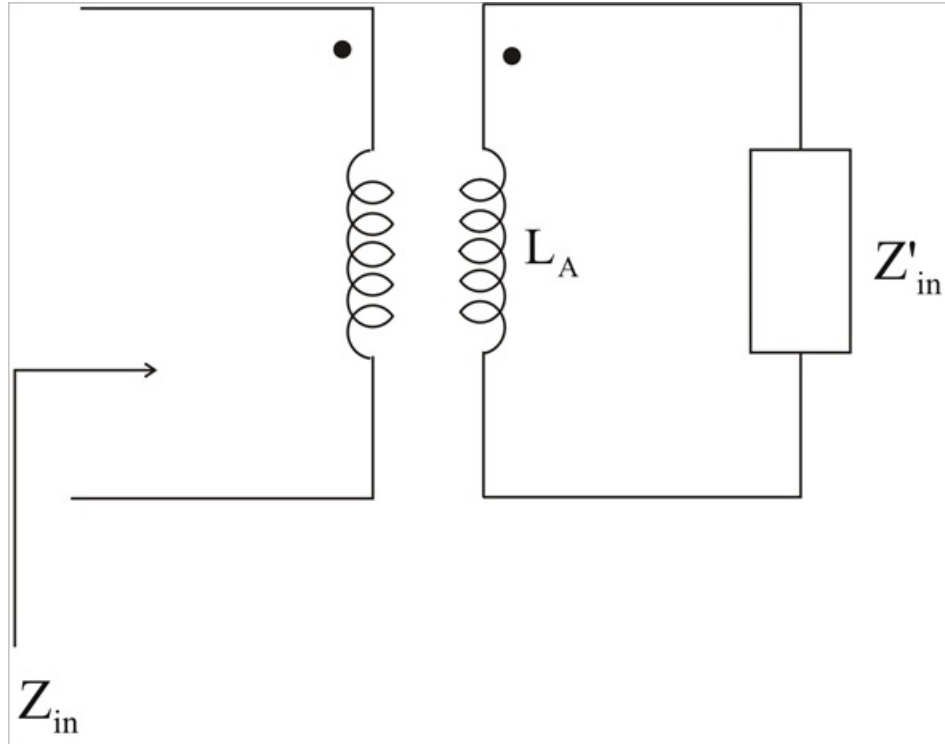


Step 3 of 5

$$\begin{aligned}
 Z'_{in} &= \frac{j\omega L_b + \omega^2 M_b^2}{(R + j\omega L_b)} \\
 &= \frac{(j\omega L_b R - \omega^2 L_b^2 + \omega^2 M_b^2)}{(R + j\omega L_b)} \rightarrow (1)
 \end{aligned}$$

For the first stage, we have the circuit below.

Step 4 of 5



Step 5 of 5

$$\begin{aligned}
 Z_{in} &= \frac{j\omega L_a + \omega^2 M_a^2}{(j\omega L_a + Z_{in})} \\
 &= \frac{(-\omega^2 L_a^2 + \omega^2 M_a^2 + j\omega L_a Z_{in})}{(Z_{in} + j\omega L_a)} \rightarrow (2)
 \end{aligned}$$

Substituting (1) into (2) gives,

$$\begin{aligned}
 &= \frac{-\omega^2 L_a^2 + \omega^2 M_a^2 + j\omega L_a \frac{(j\omega L_b R - \omega^2 L_b^2 + \omega^2 M_b^2)}{(R + j\omega L_b)}}{j\omega L_a + j\omega L_b R - \omega^2 L_b^2 + \omega^2 M_b^2} \\
 &= \frac{-R\omega^2 L_a^2 + \omega^2 M_a^2 R - j\omega^2 L_b L_a + j\omega^3 L_b M_a^2 + j\omega L_a (j\omega L_b R - \omega^2 L_b^2 + \omega^2 M_b^2)}{j\omega R L_a - \omega^2 L_a L_b + j\omega L_b R - \omega^2 L_a^2 + \omega^2 M_b^2} \\
 Z_{in} &= \frac{\omega^2 R (L_a^2 + L_a L_b - M_a^2) + j\omega^2 (L_a^2 L_b + L_a L_b^2 - L_a M_b^2 - L_b M_a^2)}{\omega^2 (L_a L_b L_b^2 - M_b^2) - j\omega R (L_a + L_b)} \\
 &= (12.764 + j7.154) \text{ Ohms} .
 \end{aligned}$$